WindsurfingMMZB — Gameplay & Control Design (v0.2 direction)

From a voice session with Tarpie, 23 October 2026. Goal: turn the player from a passenger into the sailor. Physics stays as real as possible; skill and practice must make a visible difference between new and experienced players.

Why this needs addressing

v0.1 is playable and the physics are already far along (fin flex, lift and grip, waves, gusts, wind shifts, visible water). The problem is not the simulation, it is the player's role in it.

- **The player is a passenger.** On a reach you point the gear and wait until it is time to tack. The autopilot does the sailing; the player mostly watches. The user experience feels empty.
- **Upwind is dead time.** On a course race leg you sit there pointing upwind and wait for the layline. That is the longest part of a race and currently the least interactive part of the game.
- **Practice does not pay off.** There is almost nothing separating a player who has put in hours from someone who just started. Without a skill ceiling there is no reason to keep playing or to get better.
- **No connection to the sailor.** The things a real sailor is constantly doing — committing or backing off, railing the board, absorbing chop, timing a jibe — are either automated or absent, so the player never feels like the person on the board.

The fix is not to make the simulation less real. It is to hand the player the inputs a real sailor uses, and to let the existing physics decide what happens when they get it wrong.

Principles

- The autopilot keeps sailing; the player's input sits on top of it as the sailor's body (weight, rail, trim, absorb).
- No extra HUD indicators. Feedback comes from what a real sailor sees, hears and feels: board heel and pitch, spray, sound, wave faces, dark gust patches on the water.
- All failure states come from existing physics (fin lift/grip/flex vs stall, board pitch, airtime), not scripted fails.
- Difficulty grows naturally with conditions: flat water is easy, chop and gusts demand more.

Out of scope here

- Top-speed tuning (about 50 km/h at 16–18 kn vs ~60 km/h real). Tarpie knows the fix; handled separately.

1. Reaching: push vs risk

- Player controls how hard they commit (sheet in, hang on the boom). More push = more speed, more risk of spin-out (fin stall) or catapult.
- Warning signs, all derived from physics: board chatter, more tail spray, rising pitch of the sound as fin lift approaches the stall limit; nose lift and bounce when flying too much.
- Skill: read gusts on the water, ease off just before a gust hits, push again after.

2. Chop: absorbing waves

- In waves the player must absorb (bend knees, weight forward) as the nose hits a wave face.
- Good timing: board stays in the water, fin keeps grip. Bad timing: board launches off the ramp, fin loses grip in the air, landing in spin-out or catapult.
- Becomes a rhythm game driven by real wave spacing (wind, fetch). Pushing harder = higher speed = tighter timing window.

3. Upwind: railing and flying

- Core upwind skill: the fine line between flying the board on the rail and fin (less wetted area, less drag) and losing too much speed or blowing out. Formula-style feel, not a Formula game.
- Continuous balance, not a button:
  - Too little: board flat, drags.
  - Sweet spot: windward rail lifted, thin strip of hull in the water, spray off leeward side.
  - Too much: board flies out, fin lets go, slide sideways or crash.
- Gusts: push harder to hold the rail. Lulls: ease off or the board drops down.
- Pointing vs speed trade-off and playing wind shifts (tack on headers, hold lifts) read from the water and the board, not from a VMG readout.

4. Manoeuvres

- **Tack:** keep it simple, one input to step around the front of the mast.

- **Jibe:** timed sequence of carve, sail flip and foot switch. Flip too early = stall, too late = catapult. Timing window tightens with wind strength.

5. Control scheme

One stick (or the mouse) is the sailor's body.

Input	Mouse	Controller
Rail / carve	horizontal movement	left stick X
Fore-aft trim	vertical movement	left stick Y
Absorb wave	mouse button	trigger (analogue: soft press = small chop, hard press = big ramp)
Tack / jibe steps	keys or buttons	face buttons

Design notes

- A mouse has no physical centre. Use a virtual centre that slowly springs back, so players don't drift off to one side and run out of desk.
- Original idea was absorb on a fast forward mouse flick. Risk: shares the axis with trim, so a quick trim correction could trigger an absorb. Preferred: absorb on a separate button/trigger. If the flick is kept, it must be separated from trim by velocity threshold and tested for false triggers.
- Both mouse and controller should be supported later; same mapping logic on both.

Open questions

- Exact timing windows per wind strength and skill curve.
- Spring-back speed of the virtual mouse centre.
- Whether push on the reach shares the rail axis or needs its own input.
- Race opponents (AI) vs time trial, which affects upwind tactics like dirty air.